Predicting Train Delay in the Dutch Railway Network
A Hybrid Spatio-Temporal Approach

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ABSTRACT

Accurate prediction of train delays and delay propagation can greatly reduce the amount of delay and its associated cost. Where earlier studies on delays in the Dutch railway network often focused on a subset of the network over a limited amount of time, this study incorporates the full NS (Nederlandse Spoorwegen) network over six years (2019-2024) including features such as weather data, track maintenance and passenger volume. This study also proposes a hybrid combination of GRU (Gated Recurrent Unit) and GAT (Graph Attention Network) as well as a hybrid of GAT and LSTM (Long Short-Term Memory) by replacing the linear transformations in GRU and LSTM cells with GAT for an improved integration of spatial and temporal features. The application of the GAT-GRU and GAT-LSTM in binary classification resulted in balanced accuracies of, respectively, 0.7482 and 0.7683 significantly outperforming the separate models as well as baselines in earlier work. When applying the hybrid models to the prediction of the average train delay at the next five stations, the models achieved an MAE of 0.8561 and 0.8622 minutes. The hybrid models significantly outperformed the standalone GRU and LSTM, confirming the improved performance of the hybrid model architecture.

KEYWORDS

Train Delay Prediction, Delay Propagation, Spatio-Temporal Modeling, Hybrid Deep Learning

GITHUB REPOSITORY

https://github.com/vdmaas98/thesis_repo-Jonathan_vd_Maas

1 INTRODUCTION

Train delays pose a major issue for both passengers and railway companies. These service disruptions result in longer travel times and overcrowded trains, which affects both the schedules and the overall experience of delayed travelers. In 2015, an estimated 14 million hours were lost in delayed passenger trains in the Netherlands, which, in addition to frustration, cost more than 145 million euros [30]. The high cost associated with train delays warrants research into train delay prediction. In a combined effort to increase their ability to predict short-term delays, NS (Nederlandse Spoorwegen) and ProRail set out a competition to predict short-term train delays at a maximum of twenty minutes ahead in time. At that time, the most used delay prediction strategy was that the delay at the previous station would remain unchanged [5]. In this competition in 2018, neural networks outperformed Bi-Level Random Forests and Non-Homogeneous Markov Chains [11].

Various approaches to train delay predictions have been applied in recent years, with techniques ranging from diffusion-based models to random forests and graph-based approaches to time series approaches [32]. This study builds on previous research, where the prediction of missing links through an XGBoost model was successfully applied to the Brazilian bus service and domestic airlines in the United States [17]. A similar model was applied to the Dutch railway network, but found limited success in non-simultaneous tests due to the complex nature of the network and the inability to model delay propagation through the network [14]. Non-simultaneous tests are tests in which a model is applied to a time period it was not trained on. Delays can propagate through the network due to interconnected services and stations, creating ripple effects throughout the network. This is especially a major issue in The Netherlands due to the high complexity of the railway network compared to other countries and the relatively short distances between train stations. Modern deep learning methods such as GAT (Graph Attention Network), GRU (Gated Recurrent Unit) and LSTM (Long Short-Term Memory) have been applied more successfully, however, this was done on a limited part of the railway network [9] [39]. This study builds on that by proposing a hybrid GAT-LSTM and GAT-GRU model for improved integration of spatial and temporal features. The GAT layers used in this study are GATv2 proposed by Brody et al. as an improvement on the original GAT layer [3]. To show the improved predictive capabilities of the proposed hybrid models, a standalone LSTM, GRU and GAT were trained as well and then compared to the hybrid models. Earlier efforts in delay propagation prediction in the Dutch railway network have not focused on building hybrid models and often focused on small subsets of the network instead of the network as a whole while also taking into account a less diverse feature set over a shorter time span.

The aim of this study was to develop a hybrid GAT-GRU and GAT-LSTM model for prediction of delays in the Dutch railway network. By including both temporal and spatial features such as passenger flow, weather data and network features, an improved ability to predict delay propagation was expected compared to earlier work [14]. By knowing where and when delays occur, the NS can improve the way they handle delays by, for example, optimizing crew assignment. The research questions that were answered in this study are the following:

Main research question:
To what extent can a hybrid GAT-LSTM and GAT-GRU model outperform standalone GRU, LSTM and GAT models in predicting train delay across the Dutch railway network using combined spatial and temporal data?

Sub-research questions:

- How do spatial and temporal features of the Dutch railway network contribute to the GAT attention weight distribution?
- To what extent does the increased complexity of an LSTM improve the model's predictive accuracy compared to a less complex GRU?
- How much influence does graph network complexity have on the predictive capabilities of the GAT-GRU and GAT-LSTM models?

This study contributes to the field of train delay prediction in the Dutch railway network by incorporating GAT attention mechanisms in GRU and LSTM for an improved understanding of spatial and temporal dependencies. Apart from that, this study utilizes an extensive timeseries dataset ranging six years (2019-2024), incorporating publicly available data on weather, disruptions, track maintenance, passenger volume and network structure.

2 RELATED WORK

The Dutch railway network has a very high interconnectivity where each station is connected to many others and the distances between stations are relatively short compared to the train networks in other countries [6]. Modeling challenges such as delays propagating to other trajectories arise because of this, creating the need to capture complex spatial and temporal dependencies.

2.1 General predictive transportation models

Many different methodologies have been applied to forecasting delays ranging from Bayesian networks and Markov chains to SVMs and Neural Networks [32]. Prediction accuracies up to 86,91% have been found for a few stations in The Netherlands when applying LSTM for short-term delay predictions [39] on a subset of the network.

An approach used to predict delays in the Dutch railway network adapted a method published in Nature magazine to predict missing links in the US air transport system [14]. Although the model demonstrated success in some areas, it had limited success in non-simultaneous testing. This limits the generalizability of the proposed model to unseen times. Modeling the ripple effect of delays, which is a phenomenon more prevalent in the densely interconnected Dutch railway network compared to the U.S. air network, also proved to be an issue. The previous study did not include temporal features, which are essential for capturing time-dependent dynamics [17]. The importance of adding temporal features in modeling was shown by research on the influence of track maintenance on delays in the Swedish railway network. Trains were found to be 1.43 times more likely to experience delays when passing through a part of the network with track maintenance compared to undisturbed sections. The likelihood of trains recovering from delays was also found to reduce by 11% in sections with track maintenance [12]. Similarly, research on the influence of weather on train delays showed a significant influence of temperature, snow depth and precipitation on train delays [38]. Therefore, both track maintenance and weather data were included in this study.

2.1.1 Deep learning. The complexity of the Dutch railway network warrants the use of more complex deep learning models, which was confirmed by Huang et al. in 2024 [9]. The results of their proposed model in a classification problem can be found in Figure 10 in the appendix. It clearly shows the benefit of deep learning models compared to simpler models such as Random Forest and SVR even though it greatly increases training time (FCLL-Net trained 25 times longer than SVR on the same train line). An issue with these models is that the explainability of the results usually decreases with model complexity. A GAT model was proposed to improve both the predictive performance and the explainability of the results compared to a standard GCN (Graph Convolutional Network) as the GAT outcomes at each node can be interpreted by the learned attention weights [9]. It resulted in 29% reduction of the RMSE showing the potential of using an attention mechanism in delay prediction, as it allows the model to make distinctions between the influence of different connecting nodes in the network.

2.2 Hybrid spatio-temporal models

Research on hybrid spatiotemporal models has shown promise in addressing both the spatial and temporal nature of network delays. Hybridization of two models can be done in multiple ways, such as including embeddings from one model as input in the other or, as in this study, replacing linear transformations in RNNs (Recurrent Neural Network) by GAT attention mechanisms as this makes the learned GAT weights time dependent.

One study implemented a hybrid GCN-LSTM model to predict traffic flow in New York. The experimental results demonstrated improved performance compared to competing methods, highlighting the potential of hybrid models for extracting spatio-temporal features and to capture ripple effects in delays [18]. Another study focused specifically on train delays, using a hybrid model that split data into spatio-temporal and time-series components. Their hybrid model outperformed several baseline models, however, the utilized spatio-temporal block lacked the ability to distinguish spatial and temporal features [37].

In Australia, a hybrid model that combined LSTM with CPS was used to predict train delays. This method was designed to be applied to a single service, which makes it not applicable to the current study, however, it does demonstrate the potential of hybrid models to outperform standalone LSTM models [40]. The need for more complex models was also found in similar fields such as network traffic prediction. The MAPE for LSTM, GRU and a GCN-GRU hybrid can be found in Figure 11 in the appendix, which shows the lowest MAPE for the hybrid GCN-GRU in two network topologies under different traffic intensities [41]. This shows that the additional gates in LSTM compared to GRU do not always increase model performance [42]. Therefore, both LSTM and GRU were considered in this study.

2.3 Summary and research gap

This section discussed a subset of many of the solutions to the train delay prediction problem and, added to that, hybrid solutions to delay predictions in similar fields. The discussed works show the improved performance of complex models compared to simpler models such as SVR and Random Forest. Many of the delay prediction solutions for the Dutch railway network are based on the delay prediction competition set out by ProRail and the NS in 2018 [11]. The dataset in this competition only consisted of three months of data in 2017 and the models were only tested by predicting the state of the network at three moments for each of the last four Tuesdays in the dataset. Using only this limited amount of data and evaluating on a few Tuesdays in autumn heavily limits the generalizability of the proposed solutions. To improve on this, the models proposed in this research were trained on data from 2019 to 2022, 2023 was used as validation and evaluation was done on all data from 2024. It was done to fully understand how well the models generalize to future timestamps that are not seen in training and how well the models predict delays throughout the year instead of only a specific season.

Apart from that, an area which has not received enough attention for this prediction problem in the Dutch railway network is the application of hybrid models to better incorporate both spatial and temporal features of the network. Hybrid approaches to predict train delays and similar problems such as road and network traffic delays showed improved predictive accuracy compared to standalone deep learning models [41]. The contribution of this research is therefore the training of two new hybrid models on the full NS railway network instead of a subset using a newly constructed, much more extensive time series dataset spanning six years which incorporates network features, track maintenance, disruptions, passenger volume and weather data. The construction of the dataset is expanded upon in the next section.

3 METHODOLOGY

3.1 Constructing the datasets

The datasets utilized in this study are a combination of multiple publicly available source datasets. This includes data on historic track maintenance, past hourly weather data for all the train stations in the network, passenger volume data per station, all disruptions reported by the NS and all train services including delays per station. The study focuses solely on services by the NS and not regional railway companies as no passenger volume data was available for the stations serviced by the regional railway companies.

To construct the railway network, a combination of two datasets was employed containing the geographical location of each station and geojson data of all the connections between stations. Each model was trained on the same node (station) and edge (trajectory) timeseries datasets which together form the full hourly graph network allowing for time dependent GAT attention weight updates [36].

The full node and edge timeseries were created by cross joining all stations and connections with a time index table which only consists of a consecutive, hourly timeseries. This resulted in a six year, hourly timeseries for each station and connection. For the node (stations) dataset, all source datasets were then joined to it based on station code (station name abbreviation) and time id (timestamp). In order to do this, for some features, aggregations had to be made. For example, to show how many minutes of a specific hour a station or connection was hindered by a disruption an hourly aggregation per station of this metric was made. For the edge (connection) dataset, the node level features were joined to the source and target stations of each connection through the station id which is a unique station identifier from the source stations dataset. A table with all source datasets and how they were joined to the final dataset can be found in Appendix B as well as the full pipeline executed in this study.

3.1.1 Source datasets. The basis for both datasets are the stations and connections datasets sourced from Rijden de Treinen [28], which processes real-time data from NDOV (Nationale Data Openbaar Vervoer) [20], and the NS API [33]. Apart from the name and geographical location, the stations dataset also includes a station size indication. Historical weather data was collected on an hourly basis for each station in The Netherlands based on the longitude and latitude provided in the stations dataset. The Open-Meteo Historical Weather API was used to collect the data [25]. The extracted weather data provide insights into the rain, snow depth, wind speed, wind direction and temperature at each train station. The temperature was split up into soil temperature 0 - 7 cm below ground and the temperature two meters above ground, as soil temperature influences soil strength and therefore the stability of the tracks [8] and above ground temperature influences track deformation through thermal expansion [35]. Open-Meteo provides API access to many different validated weather datasets such as ECMWF IFS, ERA5, and ERA5-Land [29].

Both the disruptions and services data was publicly available on Rijden de Treinen. The maintenance data was sourced from Rijden de Treinen as well, however, this data was not yet downloadable, so it was scraped from the site. Prior to this it was checked whether any licenses or robot.txt files prohibited this which was not the case. The maintenance data includes all announced maintenance operations spanning the full six years of the dataset. The last feature, which was included in the dataset was the number of passengers boarding and deboarding trains per station, as well as the number of passengers transferring at each station. This data was made available by the NS up until 2023 [22]. Since the year-end report of the NS for 2024 states that the total passenger volume increased by 4.3% [21] the data for 2023 was increased by this number to synthesize passenger volume data for 2024. To quantify the influence of this introduced bias on model performance, three logistic regression models were trained. Each had a mean 4.3% passenger volume increase across all stations, however, for two models the distribution over stations was done randomly with passenger volume increases between 0% and 10%. All three models achieved the same balanced accuracy, showing the negligible influence of this introduced bias.

3.1.2 Feature engineering. Since the node and edge datasets also contained rows without services and therefore no delay, features were engineered for each to signify whether a node or edge had services at timestamp t. Training and evaluation was only done on these rows in order not to artificially improve the results as nodes and edges without services are never delayed, which simplifies prediction.

In classification, the target was defined as a trajectory-timestamp combination having a higher ratio of delayed trains than the total ratio of delayed trains in the full dataset ($is\_significantly\_delayed$). This definition was chosen to approach a balanced dataset. It resulted in a 40/60 class balance for the edge-timestamp combinations with services. To account for the class imbalance in the loss calculation, the pos_weight was set to 1.5 (60/40). To also take delay propagation into account, the regression target features were $average\_next\_station\_x\_delay$ ($x \in \{1, 5\}$). This meant that from station y, the average delay at station x is the average delay of trains leaving station y at timestamp t and traveling an x number of stations along valid train services specified in the source services dataset. For regression, another mask was applied to the dataset again to not artificially improve the test results. If, from node y, the $next\_station\_x\_count$ for any station x was zero, this meant that the corresponding $average\_next\_station\_x\_delay$ had to be an artificially added zero as a result of data preprocessing and not an actual zero minute delay. These cases were also masked to completely mask any artificially added zero delays from the target features.

Great care was taken not to introduce any features that could not be known at the time of prediction. The only future features introduced in the data were planned maintenance activities. Just as in the classification problem, the models were always trained to predict the state of the system at time $t + 1$ where the state of the system up to time t was known with t in hours. This often stretches the prediction horizon much further than in previous works. An overview of all features and the corresponding definitions used in the modeling can be found in Appendix G.

3.2 Exploratory data analysis

Exploratory data analysis (EDA) was done to find patterns between weather, track maintenance, passenger volume and delays. Initially, the different source datasets described above were investigated further to check for missing data and duplicate rows. This revealed no major issues in the data, as only the arrival and departure times showed about 10% missing values, which was expected as a result of trains having an origin and terminus station and therefore no departure delay after a service has reached its final destination. The mean and total delay for 2023 and 2024 were found to be about 1.8 times higher than in the four years before which. Plotting the distribution of the maximum delays revealed a heavily right-skewed distribution which was expected for this type of feature.

Next, the EDA focused on plotting delay correlations and visualizing delay propagation. Figure 1 shows a clear seasonal trend in mean arrival delays, as well as an increased mean arrival delays during rush hours. The highest mean arrival delays were found in the middle of the night, however, these are less relevant for this study due to the very low number of train services. These correlations were also found in previous works [38] [19]. Since only yearly passenger volume data was publicly available, the yearly data was distributed over each hour similar to the total arrival delay which can be found in Figure 14 in the appendix, creating a bias in the dataset.

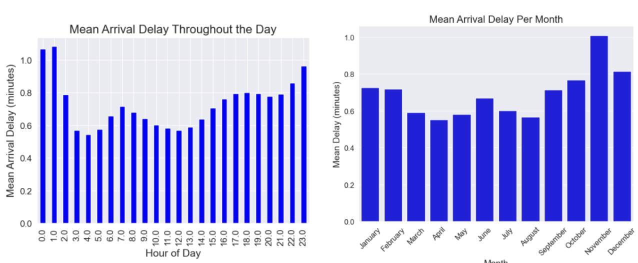

Figure 1: Distribution of mean arrival delays throughout the day (left) and throughout the year (right).

The influence of passenger volume on train delays was confirmed by plotting the mean arrival delay per station per year in Figure 2, which shows increasing mean delays for increasing passenger volume. The seasonal trend in delays was confirmed by plotting the influence of rain and wind on delays which can be found in Figure 16 and 15 in the appendix. It does show that both wind and rain have a non-zero threshold where below it no major influence on delays was found.

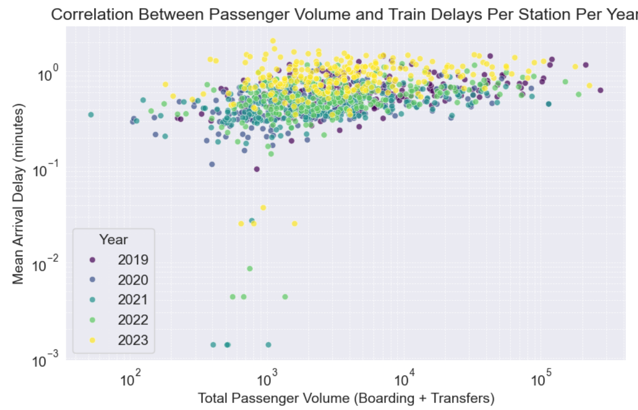

Figure 2: The correlation of passenger volume and mean arrival delays per station.

The "ripple effect" of train delays described in earlier work [14] was found in the delay data by plotting the arrival versus the departure delay for specific services, which is shown in Figure 3. The diagonal line shows services where the delays remained the same between arrival and departure at a station and the delay thus propagated through the network. It also shows that assuming departure and arrival delays remain unchanged throughout a service is not an accurate strategy for delay prediction, as a third of all services fall outside of this diagonal line.

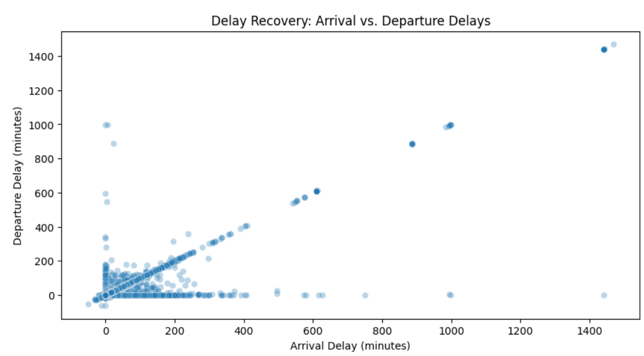

Figure 3: The arrival and departure delay for the same train service.

The initial plots of the influence of track maintenance on delays resulted in a Pearson correlation of -0.0224, which was unexpected as earlier work found track maintenance to increase delays [13]. This revealed an issue with the data, namely that only the start and end station of a section affected by maintenance was included in the data. Using the previously constructed railway network, all intermediate stations of the affected railway sections were found. The plot of the correlation between maintenance and delays at a station level revealed that it can have a large impact on train delays as shown in Figure 4. For example, for Amsterdam Bijlmer ArenA the average delay per service within a 24 hour window of planned maintenance was found to increase by 73.9%. Surprisingly, some stations also showed an increase in delay after maintenance ended, suggesting there might be lingering effects from track maintenance. Combined with previous works outlined in Section 2, the EDA formed the basis for the different feature types included in the final dataset.

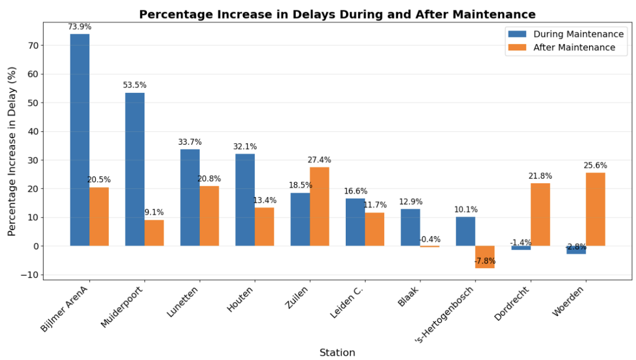

Figure 4: The percentage change in average delay per service within a 24 hour time window of a maintenance event and the time after it compared to a non-maintenance baseline.

3.3 Feature normalization and selection

To further prepare the data for model training, feature normalization and selection were performed. Different normalization techniques were applied based on the characteristics of each feature. Cyclical features such as hour, day of week, and month were transformed using sine and cosine functions to preserve their periodic nature [2]. Continuous features with Gaussian distributions, such as temperature, were standardized using z-score normalization. Passenger volume features were scaled to the [0,1] range using min-max scaling, while features prone to outliers such as maximum delays were first log normalized and then scaled using robust scaling to minimize extreme values dominating learning [24].

As a final step, feature selection was applied to the datasets. This was done by training a random forest regressor with the next station's arrival delay as a target variable. The features were ranked by importance to confirm that no features caused leakage from the future. The selected features covered temporal patterns, operational metrics, and network characteristics, ensuring a comprehensive representation of the factors that influence delay propagation [26].

3.4 Model selection and training

Earlier works outlined in Section 2 show the complexity associated with train delay prediction. This is especially the case in The Netherlands due to the high interconnectivity of the stations and the relatively small distances between stations compared to many other countries. Therefore, the model architecture proposed in this study leverages a GAT to capture the complex spatial relationships between stations, allowing the model to learn the structure of the railway network and the way delays propagate through it. GAT is a special type of Graph Neural Network (GNN) which uses attention mechanisms to weight contributions of neighboring stations whereas other GNNs such as GCN use mean aggregation [34]. Previous research has shown that not each station contributes equally to the delay, making GAT a suitable choice for this study [9]. The temporal dependencies in train delays were modeled using both LSTM and GRU, ensuring that the recurrent patterns and fluctuations in the data are effectively captured. Both architectures were explored to assess whether the reduced computational cost of the GRU outweighs potential performance differences compared to the LSTM. Transformer based models were not considered here as they require O(N2̂) memory, making them inefficient for an hourly timeseries dataset that spans multiple years [7]. This is less of an issue for GAT since attention is based on the connected stations instead of the entire network. The forward pass for GRU, LSTM and GAT used in this study can be found in Appendix F.

3.4.1 Model architecture. To fully capture spatial and temporal dependencies, a hybrid model architecture was explored where alterations to the RNNs were made similar to the DCRNN architecture that was previously applied to traffic forecasting [10]. In that study, the linear transformations in the GRU gates were replaced by diffusion convolutions. In this study, the linear transformations in the GRU and LSTM gates were replaced by GAT transformations. At each timestep t, the input consists of: node features $x_t \in \mathbb{R}^{n \times f_n}$, where n is the number of stations and f_n is the node feature dimension, graph structure e_t representing the valid connections between stations and edge features $a_t \in \mathbb{R}^{e \times f_e}$ where e is the number of edges and f_e is the edge feature dimension as well as the previous hidden state h_{t-1}.

GAT-GRU modifies the GRU forward pass as follows:

$$
\begin{aligned}
r_t &= \sigma\left(\mathrm{GAT}_r(x_t, e_t, a_t) + \mathrm{GAT}_r(h_{t-1}, e_t, a_t)\right) \\
u_t &= \sigma\left(\mathrm{GAT}_u(x_t, e_t, a_t) + \mathrm{GAT}_u(h_{t-1}, e_t, a_t)\right) \\
\tilde{h}_t &= \tanh\left(\mathrm{GAT}_c(x_t, e_t, a_t) + r_t \odot \mathrm{GAT}_c(h_{t-1}, e_t, a_t)\right) \\
h_t &= u_t \odot h_{t-1} + (1 - u_t) \odot \tilde{h}_t
\end{aligned}
$$

GAT-LSTM modifies the LSTM forward pass as follows:

$$
\begin{aligned}
i_t &= \sigma\left(\mathrm{GAT}_i(x_t, e_t, a_t) + \mathrm{GAT}_i(h_{t-1}, e_t, a_t)\right) \\
f_t &= \sigma\left(\mathrm{GAT}_f(x_t, e_t, a_t) + \mathrm{GAT}_f(h_{t-1}, e_t, a_t)\right) \\
o_t &= \sigma\left(\mathrm{GAT}_o(x_t, e_t, a_t) + \mathrm{GAT}_o(h_{t-1}, e_t, a_t)\right) \\
g_t &= \tanh\left(\mathrm{GAT}_g(x_t, e_t, a_t) + \mathrm{GAT}_g(h_{t-1}, e_t, a_t)\right) \\
c_t &= f_t \odot c_{t-1} + i_t \odot g_t \\
h_t &= o_t \odot \tanh(c_t)
\end{aligned}
$$

Additional layer normalization was applied at the end of both cells to stabilize training. To make edge predictions for link prediction, the edge attributes were used as extra input for the final layer, as well as the source and target node features. The layer consists of two linear transformations and GELU activation. Since BCEWithlogitsloss was used as the loss function, no extra sigmoid layer was needed. For the final layer in regression, only the current node features were used to make predictions with L1Loss as the loss function, as it directly calculates the MAE. In evaluation, the true and predicted delays were first unscaled before MAE and RMSE was calculated. The hybrid model architecture can be found in Figure 5.

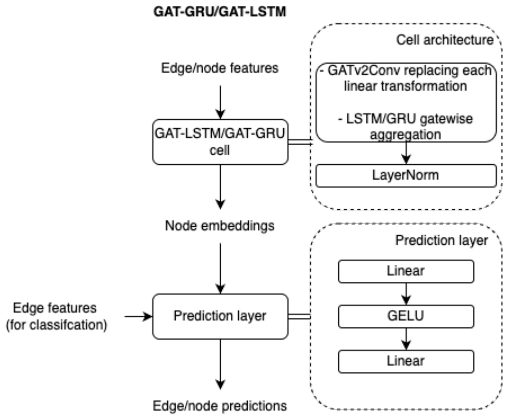

Figure 5: Overview of the hybrid model architecture for both the GAT-LSTM and GAT-GRU.

3.4.2 Training. Initially, a hybrid model architecture where static attention weights were introduced as extra feature in the GRU and LSTM was also considered, however, the time-dependent attention weights were expected to be more beneficial to the hybrid model performance. The attention weight computations involve only immediate neighbors in the railway network, which makes the model computationally more efficient compared to attention over the entire network as in transformer based models. For each model, an early stopping patience of ten epochs was used as well as a dropout of 0.3 in the forward pass to prevent overfitting. Apart from that, L2 regularization was applied through weight decay and for each epoch the model was only saved if the validation score improved. The latter causes weight updates in epochs after the last validation score improvement to be discarded, leading to better generalization to new data. Each model containing GAT had two attention heads to capture more complex attention patterns. The rest of the model hyperparameters can be found in Table 6 in the appendix.

3.5 Validation and evaluation

Validation and testing involved the use of a validation and test set to assess the generalization capabilities of the models. Every model was trained on all the data from 2019 to 2022 to incorporate both data from before and during the COVID-19 crisis, making it more robust to changes in the data. For the validation set, all data from 2023 was used and each model was tested on the full year of 2024. By making a time-based train-test split, the model was evaluated on timestamps that were unseen during training, whereas a random split could introduce data leakage.

3.5.1 Baselines. The performance of the hybrid models was compared to multiple baselines. For the regression problem, the mean delay of the train set was used for every prediction as a baseline to see if the models actually found useful generalizable patterns in the data. For the classification problem, a logistic regression model was trained. The same train-test split and masking strategies were applied to these baselines for a fair comparison. Since this study builds on the link prediction model proposed by Lei et al. [17], the model performance was compared to this model based on balanced accuracy. The models were also compared to an LSTM trained on the Dutch railway network [39] in MAE and RMSE. This comparison was also made with respect to the separate LSTM, GAT and GRU models to show the change in performance of the proposed hybrid models compared to their separate counterparts applied to the same dataset. To further show the ability of the hybrid models to forecast delay propagation, they were constructed with multiple outputs for the regression problem. Therefore, not only the arrival delay at the next station, but also at stations further down the line were predicted similar to earlier work [1].

3.5.2 Feature importance. In addition to predicting delays, feature importance analysis was done as well to improve the explainability of the proposed models, similar to feature importance analysis by Huang et al. [9]. It was done by extracting the GAT attention weights of the test set at each timestep. Spearman correlation between each feature and the weights was calculated. The attention weights for all edges were extracted for both actual trajectories and the self-loops introduced by GATv2Conv separately and then divided between peak and off-peak hours. This resulted in four Spearman correlations for each feature group per hybrid model (total $n = 8$). The statistically significant correlations were then semantically grouped (weather, service, network, etc.) and the Wilcoxon signed-rank test was used to rank feature groups. The Wilcoxon signed-rank test was chosen here instead of a pairwise t-test, as with a sample size of $n = 8$ it is difficult to assume that normality holds [15].

4 RESULTS

In both the regression and classification problem, the model architectures were kept as similar as possible to properly evaluate the predictive capacity of the different models. For both problems, bootstrap sampling was used to calculate the 95% confidence interval in Table 1 and 3. The only alterations that were made when retraining the models for regression was that in classification the edge features were inserted again in the final layer to make a single edge-level prediction whereas in regression five node-level predictions were made and the edge attributes were not inserted again. Significance in this section is defined as test results with $p < 0.05$. The output tables of the statistical validation of the results can be found in Appendix H.

4.1 Prediction of significant delays

The first application of the proposed models required to predict the binary flag $is\_significantly\_delayed$. The results can be found in Table 1. As expected, the table shows an increasing balanced accuracy with model complexity with very similar results for the separate GRU, LSTM and GAT. Pairwise t-tests using the monthly balanced accuracy from Figure 6 revealed that the GAT-LSTM significantly outperforms the other models. No significant performance increase was found for GAT-GRU compared to GAT. The t-test results can be found in Table 7. The proposed hybrid models offer significant improvements in non-simultaneous testing compared to the XGBoost applied to the Dutch network [14]. Important to note here is that the balanced accuracy of 0.82 was the result of applying XGBoost to the US air traffic network, which is less complex than the Dutch railway network due to longer distances between nodes.

Model	Balanced Accuracy
Logistic Regression	0.6571
GRU	0.6927 [0.6499, 0.7353]
LSTM	0.7001 [0.6594, 0.7391]
GAT	0.7216 [0.6841, 0.7589]
GAT-GRU	0.7482 [0.7376, 0.7576]
GAT-LSTM	0.7683 [0.7594, 0.7765]
XGBoost (Dutch railway) [14]	0.5400
XGBoost (US Air) [17]	**0.8200**

Table 1: Test balanced accuracy for prediction at time t of whether an edge was significantly delayed at time $t+1$ in a non-simultaneous testing scenario.

In addition to the comparison of the balanced accuracy for the full test set, the accuracy per month was also plotted and can be found in Figure 6. This figure shows that even though the balanced accuracies over the full year are significantly higher for the hybrid models than for the separate counterparts, the highest balanced accuracies per month are not. The main improvement which the figure shows is that where the GRU, LSTM and GAT have a clear seasonal trend in their balanced accuracies, the hybrid models achieve a much more stable balanced accuracy throughout the year. A direct result of this is the narrower confidence intervals for the hybrid models in Table 1.

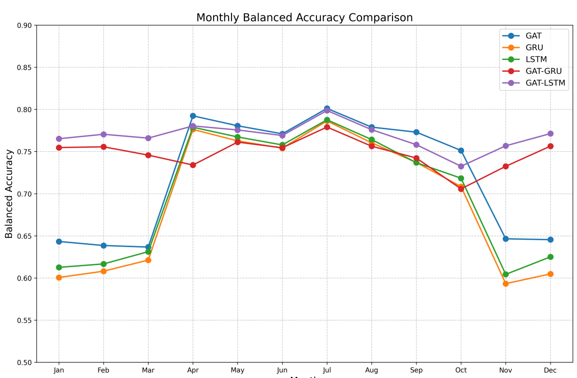

Figure 6: Comparison of the monthly balanced accuracies for each of the five models trained in this study.

Figure 7 shows the balanced accuracy per trajectory throughout the entire network for GAT-LSTM. The grey edges signify that there are no valid services in the dataset for that trajectory as they are serviced by regional railway companies. The figure shows decreasing balanced accuracies for the most complex parts of the network. This is supported by the linear regression coefficients from Table 10 which show that especially the node degree of the target station has a significant negative influence on the balanced accuracy.

To improve the explainability of the model, an analysis of the GAT attention weights was done as outlined in the methodology.

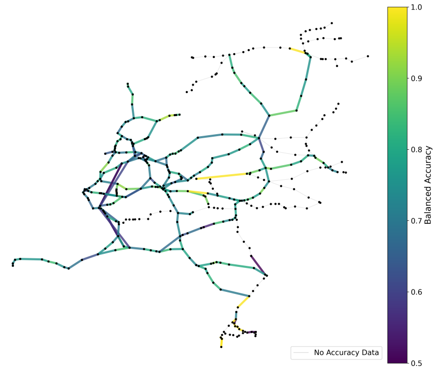

Figure 7: GAT-LSTM balanced accuracy for each edge in the railway network. Grey edges indicate links without services in the dataset.

The mean of the absolute Spearman correlations per feature group is reported for significant correlations in Table 2. The Wilcoxon signed-rank test for each feature group revealed a group of three most important feature clusters (service, network and passenger volume), which were significantly more important than the other four but no significance claims could be made when comparing them to each other. The results of this test can be found in Figure 17 in the appendix. The test results also showed maintenance features to be more important than weather, with delay history, temporal features and weather being the least important.

Model	Temp	Net	Weath	Pass	Serv	Delay	Maint
GAT-GRU Peak	–	0.462	0.120	0.574	0.479	0.455	0.283
GAT-GRU Off	–	0.458	–	0.600	0.502	0.506	0.291
GAT-LSTM Peak	–	0.205	–	0.198	0.162	0.115	0.165
GAT-LSTM Off	–	0.196	–	0.182	0.248	–	0.151
GAT-GRU Peak (S)	0.134	0.252	–	0.213	0.129	–	–
GAT-GRU Off (S)	0.099	0.243	–	0.236	0.121	–	–
GAT-LSTM Peak (S)	–	0.244	–	0.106	0.142	–	–
GAT-LSTM Off (S)	–	0.238	–	0.107	0.142	–	–

Table 2: Spearman correlations between attention weights and feature groups. Temp = Temporal, Net = Network, Weath = Weather, Pass = Passenger Volume, Serv = Service, Delay = Delay History, Maint = Maintenance. (S) denotes self-loops and – means no significant correlations were found.

4.2 Forecasting delay magnitudes

Where the classification problem focused on predicting whether trajectories are significantly delayed or not, the regression task focused on predicting the magnitude of the delays for each station in the network. The average MAE and RMSE of the different models can be found in Table 3 including the 95% confidence interval.

The results again show the increased performance by both hybrid models, confirming the importance of both spatial and temporal features in the models. The t-test results in Table 8 and Table 9 show that the GAT-GRU significantly outperformed all other models, except for the GAT, in MAE where no significance claim could be made due to more inconsistent monthly predictions by the GAT

Model	Average MAE	Average RMSE
Mean baseline	1.8705	2.6672
GRU	0.9335 [0.7986, 1.0972]	1.9522 [1.7983, 2.1210]
LSTM	1.1206 [1.0099, 1.2456]	1.9936 [1.9023, 2.0900]
GAT	1.0698 [0.8089, 1.3698]	2.1476 [1.7931, 2.5205]
GAT-GRU	0.8561 [0.7192, 1.0263]	1.8652 [1.6898, 2.0798]
GAT-LSTM	0.8622 [0.7273, 1.0343]	1.8652 [1.6924, 2.0676]
LSTM baseline [39]	**0.2683**	**0.4955**

Table 3: MAE and RMSE averaged across all five target stations.

Model	Temp	Net	Weath	Pass	Serv	Delay	Maint
GAT-GRU Peak	–	0.535	0.142	0.514	0.403	0.394	0.271
GAT-GRU Off	–	0.530	0.122	0.529	0.402	0.416	0.276
GAT-LSTM Peak	–	0.532	0.133	0.442	0.357	0.354	0.222
GAT-LSTM Off	–	0.549	0.128	0.473	0.371	0.386	0.240
GAT-GRU Peak (S)	0.122	0.153	0.116	0.153	0.162	0.133	–
GAT-GRU Off (S)	0.125	0.147	0.105	0.125	0.141	0.121	–
GAT-LSTM Peak (S)	–	0.125	0.128	0.178	0.183	0.192	–
GAT-LSTM Off (S)	0.100	0.115	0.130	0.164	0.187	0.197	–

Table 4: Spearman correlations between attention weights and feature groups. The same abbreviations as in Table 2 apply.

compared to the other models. For the RMSE, the hybrid models were only found to significantly outperform the standalone GRU. Again, the monthly metrics were used as samples for the pairwise t-tests ($n = 12$). The LSTM baseline set out by Wen et al. significantly outperformed all the other models, however, the long prediction horizon with delays often being predicted hours in advance still is an improvement over Wen et al. where delays were always predicted for the next train.

Figure 8 shows the average MAE for the GAT-GRU throughout the network. Again, the model performance decreases with increasing network complexity, which was confirmed by the results in Table 11. It shows that the MAE increases with increasing node degree. The average degree of the target nodes was not found to have a significant influence on the MAE, whereas it was important in the balanced accuracy showing the difference between node and edge-level predictions.

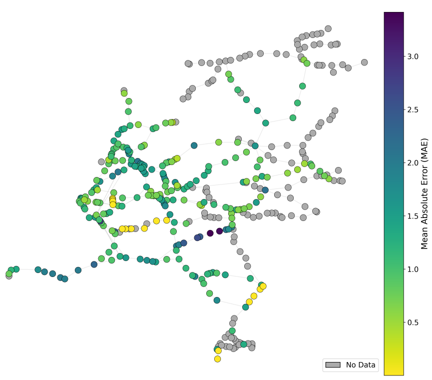

Figure 8: GAT-GRU MAE per node in the network. Grey nodes have no valid services in the dataset.

The same feature importance analysis that was done for the classification problem was also done for the regression problem. The mean Spearman correlations for each feature group can be found in Table 4 and the results of the Wilcoxon signed-rank test in Figure 18 in the appendix. This showed that for the regression problem the delay history was much more important than in classification. It is a result of the time frame of a single prediction being much longer, as a train traveling for five stations can take multiple hours, so a single prediction can be much further into the future than for the classification problem. Weather features were found to be significantly more important than temporal features but still similarly important as maintenance features.

The evaluation metrics for the regression problem were also plotted per month and are shown in Figure 9. For both hybrid models, a more clear seasonal trend is visible than in classification, which was confirmed by weather-related features having a more significant influence on the GAT attention weight distribution in regression than in classification. The figure shows the importance of delay history in the regression problem as each model's performance resembles the monthly delay distribution from Figure 1 which was not the case in classification.

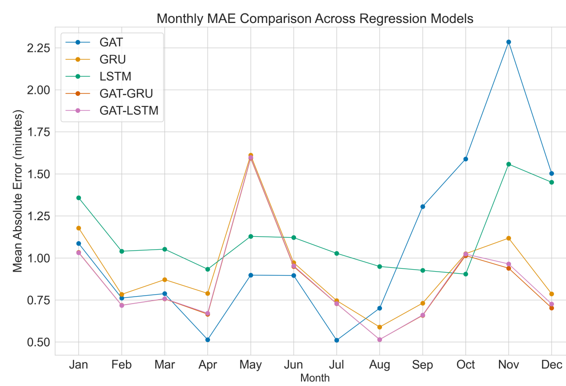

Figure 9: Temporal comparison of the MAE for each of the five models trained in this study.

5 DISCUSSION

5.1 Model performance

In this study, hybrid GAT-GRU and GAT-LSTM models were trained to predict train delays in the Dutch railway network specifically focusing on train services by the NS. The basis for this work was the link prediction study done on the US air network and the application of that model to the Dutch railway network [17] [14]. The developed hybrid GAT-GRU and GAT-LSTM were applied to both a binary classification and a regression problem to test the predictive capabilities of the models in predicting a trajectory to be significantly delayed in the next hour, as well as forecasting exact delay magnitudes. To be able to compare model performance to earlier work, the balanced accuracy was used as the performance metric for classification and both MAE and RMSE for regression. There was no comparison made to simultaneous testing results as only non-simultaneous tests were done in this study for a better understanding of the models' generalizability to unseen timestamps.

5.1.1 Overall model performance. For the GAT-GRU and GAT-LSTM hybrids, this resulted in balanced accuracies of respectively 0.7482 and 0.7683 outperforming the linear regression baseline and previous link prediction performed on the Dutch railway network at balanced accuracies of respectively 0.6571 and 0.54. The latter is an implementation of the XGBoost model proposed by Lei et al. [17]. for the Dutch railway network making the comparison to the balanced accuracy of 0.54 more accurate than 0.82 as this was achieved when applying the XGBoost to the US air network which is much less complex than the Dutch railway network due to longer distances between nodes.

In classification, the GAT-LSTM significantly outperformed all other models including GAT-GRU whereas in regression, the GAT-GRU outperformed the GAT-LSTM. Neither of the proposed hybrid models beat the baseline set out by Wen et al. [39] as a result of the current regression task being more complex. The main difference between this study and the baseline is that this study takes the full NS network into account, which are 254 stations, whereas the baseline purely focused on the railway section from Rotterdam to Dordrecht containing seven stations on a single railway line. Apart from that, they used the data of the previous train to predict the next one, whereas in this study the average delay of each of the next five stations for time $t + 1$ was predicted at time t, which is a significantly longer prediction horizon. Doing delay prediction on a single line allows for prediction of delays for specific services, whereas doing time series delay prediction on the full network meant mean aggregating delays for each hour. This alters the nature of the prediction task. A more direct model comparison between the baseline set by Wen et al. was difficult, as they only applied their LSTM to a single train line. Applying the hybrid models to a single train line does not leverage the GAT attention mechanism in the hybrid cells efficiently, as the attention weights are only distributed over the previous and next station instead of all connecting stations.

Both hybrid models significantly outperform the LSTM trained for the regression task on average MAE in this study, confirming the improved predictive capabilities of the hybrid models. The limited sample size ($n = 12$) presents a challenge when it comes to significance testing, as the small sample size reduces the statistical power of the tests, making it more difficult to establish a significant performance difference even though the mean MAE and RMSE indicate it. A future improvement would be to make the node level predictions for the GRU, LSTM and GAT as well and doing the significance tests on that data ($n = 254$).

5.1.2 Monthly performance and feature importance. The separate GRU, LSTM and GAT clearly showed a seasonal trend in balanced accuracy where a sharp increase in balanced accuracy was seen from March to April with a slightly more gradual decrease again from July until November. This follows the yearly delay pattern seen in Figure 1 where delays increase in the months where the balanced accuracy is the lowest. This seasonal trend was not found for GAT-GRU and GAT-LSTM and was confirmed by the fact that the feature importance analysis for both models found weather-related features to be among the least influential on the GAT attention weight distribution. The MAE and RMSE throughout the year does show a slight seasonal trend for the hybrid models, which was confirmed by the feature importance analysis, as weather-related features were found to be more important than in classification. Apart from that, the temporal features showed very little importance in the attention weights analysis. The reason for this is that the attention weights were extracted per hour and then averaged. Since an hour is the shortest unit of time in the dataset, very little temporal signal per weight was expected, which was then further diluted by averaging over many weights.

The rest of the feature importance analysis revealed similar patterns for both classification and regression. No significant differences were found between peak and off-peak hours. Network, service and passenger volume features were found to dominate the distribution of attention weights. A major difference between the regression and the classification task was that in regression, the delay history also dominated the distribution of attention weights. It was a result of the longer prediction horizon of the regression problem compared to the classification problem. It is also evident in Figure 9, as the monthly MAE follows a pattern similar to the monthly delay distribution in Figure 1. A linear regression analysis of the model features and GAT attention weights was initially also done, however, it resulted in fewer significant results suggesting that the GAT attention mechanism is mainly influenced by non-linear relationships.

5.1.3 Full network performance. Linear regression based on the balanced accuracy for each edge and node MAE for GAT-LSTM and GAT-GRU confirmed the pattern in the figures, where lower predictive power was observed in the most complex parts of the system. It was expected, as delay patterns are most complex in these parts of the network (Figure 7 and 8).

5.2 Limitations

As can be seen in the previous section of this paper, the proposed hybrid models show promising results for both the classification and the regression prediction problems, however, there are a few limitations that are worth discussing.

Initially, it appeared that overfitting happened in most models as a performance gap between training and validation was observed. Applying more regularization by increasing the weight decay and dropout did not change this. In EDA, both total and mean delay were found to be about 1.8 times higher for the validation and test set than for the training set, which is partly due to COVID-19, where there were less trains, less passengers and less delays [16]. Since regularization did not solve the issue, the discrepancy between training and validation accuracy appears to be a result of prediction being more difficult rather than overfitting.

The applied normalization strategy turned out to be inadequate for some features as a result of the heavily right-skewed distributions of train delays. To reduce computation time, the scalers were fitted on a random subset of approximately 25% of the data. This resulted in some heavy outliers in the other 75% not being properly scaled and possibly dominating learning [23]. To properly rescale the predictions to calculate MAE and RMSE, winsorization had to be applied to cap these remaining few outliers [31].

Figure 13 shows the Spearman correlation matrix between the different feature clusters. The high correlation between service features and both delay and passenger volume features shows that the dataset could have benefited from stricter feature selection.

The next point worth mentioning is on the generalizability of the models. The train-test split was made time-based to see if the model generalizes well to unseen times. The models were not tested on unseen stations as this would break the graph structure needed for the GAT and both hybrid models to function.

5.3 Future work

To further explore the capabilities of the proposed model architectures, forecasted features such as passenger volumes and weather should also be included in the dataset. Great care should be taken to ensure that no data leakage is introduced in the dataset as a result of adding these features. To achieve this, forecasted features for time $t+1$ should be added to the data for time t. The datasets should be further improved by applying the scalers on the full dataset instead of a 25% subset so winsorization is not needed. When 2024 passenger volume data is released, the bias of synthesizing the data can be removed. The hourly distribution of passenger volumes is not publicly available and therefore this bias remains in the dataset.

Apart from that, the model architecture can be improved as well, as currently the architecture only contains one pass of the proposed hybrid cells and then an output layer. Multiple hybrid layers or another fully connected layer before the output potentially benefits the predictive capacity of the models. This does pose challenges in terms of scalability of the models, as training times of the hybrid models are already 20-30 minutes per epoch and will increase further when increasing model complexity.

Generalizibility to unseen stations should be further tested by applying the hybrid models to railway networks of other countries or by including Dutch stations serviced by regional railway companies in the dataset. To further explore the applicability of the models to a real-world scenario, it must be tested how well the models predict large and zero delays.

From an ethical perspective, this study only used aggregated passenger volumes without any personally identifiable information included. If any new data sources were to be added to the model, it is crucial to keep in mind that this should remain the case to prevent unwanted biases from coming into the model. The dataset is intended to be made available for future use.

6 CONCLUSION

In the field of train delay prediction, this study aimed to improve the integration of spatial and temporal features, by developing a hybrid GAT-GRU and GAT-LSTM model architecture. It builds on earlier link prediction research in the US airline network and its application to the Dutch railway network. The hybrid models were constructed by replacing the linear transformations in GRU and LSTM by GAT attention mechanisms similar to how diffusion convolutions are included in DCRNN architecture. Training of the hybrid models included the use of timeseries datasets spanning six years (2019-2024) created based on publicly available data. The models were applied to both a binary classification and a regression task. In the former, a target feature showing whether the percentage of delayed trains for each trajectory-hour combination was more than that of the full dataset was used. In the latter, the average delay for x stations after the current one ($x \in \{1, 5\}$) along valid train services was forecasted. To improve the explainability of the proposed models, feature importance analysis based on GAT attention weights was done. At every prediction for time $t+1$ the models were exposed to features up until time t.

The GAT-GRU and GAT-LSTM achieved balanced accuracies of respectively 0.7482 and 0.7683 with the latter significantly outperforming the GAT-GRU and separate GRU, LSTM and GAT as well as the XGBoost baseline applied to the Dutch railway network. The main improvement of the hybrid models over their separate counterparts was that there were no seasonal performance drops in autumn and winter. In regression, the GAT-GRU outperformed the GAT-LSTM, however, neither was able to beat the LSTM baseline set by Wen et al. as a result of including the full network over six years instead of a single train line over the course of three months. The main improvement here is that the hybrid models were able to predict delays much further in advance than in the baseline work. Both hybrid models also outperformed the standalone RNN's applied to the same task. The fact that GAT-LSTM achieved the highest balanced accuracy in classification and GAT-GRU the lowest MAE in regression shows that the extra gate in LSTM offers clear benefits depending on the prediction task. For both classification and regression, the hybrid models showed a decreasing performance with increasing network complexity and feature importance analysis revealed service, network and passenger volume features to have the most influence on the GAT attention weight distribution in both problems. Delay history also played a major role in regression due to the extended prediction horizon.

Generalizability of each model to unseen time was tested by performing a time-based train-test split, where evaluation was done on a full year of unseen data. However, it was not tested whether the models generalize well to unseen stations, as this would break the graph network. To test this in a future study, the models would need to be applied to the railway network of another country. In addition to that, the regression task focused on predicting exact delays reported in MAE and RMSE to compare it to earlier work. However, more research is needed to measure how well the hybrid models predict large and zero delays to test their applicability to real-world scenarios. In conclusion, even though more research is needed, both the proposed GAT-GRU and GAT-LSTM models have shown improvements compared to previous works in the field of train delay prediction in the Dutch railway network.

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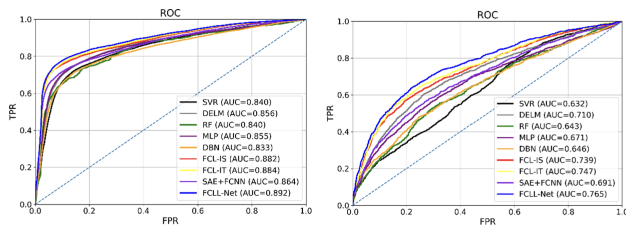
A RELATED WORK



(a) (b)

Figure 10: Comparison of MAE and MAPE for different models when applied to train delay prediction on two train lines in China [9].

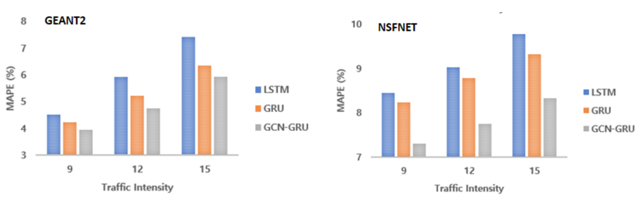


Figure 11: Performance comparison of LSTM, GRU and GCN-GRU for Link State Prediction in GEANT2 and NSFNET topologies under different network traffic intensities [41].

B FULL PROJECT PIPELINE

Dataset	Source	Join
Time index	-	-
Stations	Rijden de Treinen [28]	Cross join to time index table
Connections	NS API [33]	Cross join to time index table
Weather	OpenMeteo [25]	Join to stations on latitude, longitude and time_id
Services	Rijden de Treinen [28]	Join to stations on station_code and time_id
Planned maintenance	Rijden de Treinen [28]	Join to stations on station_code and time_id
Passenger volume	NS dashboard [22]	Join Join to stations on station_code and time_id
Disruptions	Rijden de Treinen [28]	Join to stations on station_code and time_id

Table 5: The source datasets and joins used to create both the node and edge datasets. The edge dataset was created by joining the station features to the source and target station of each edge.

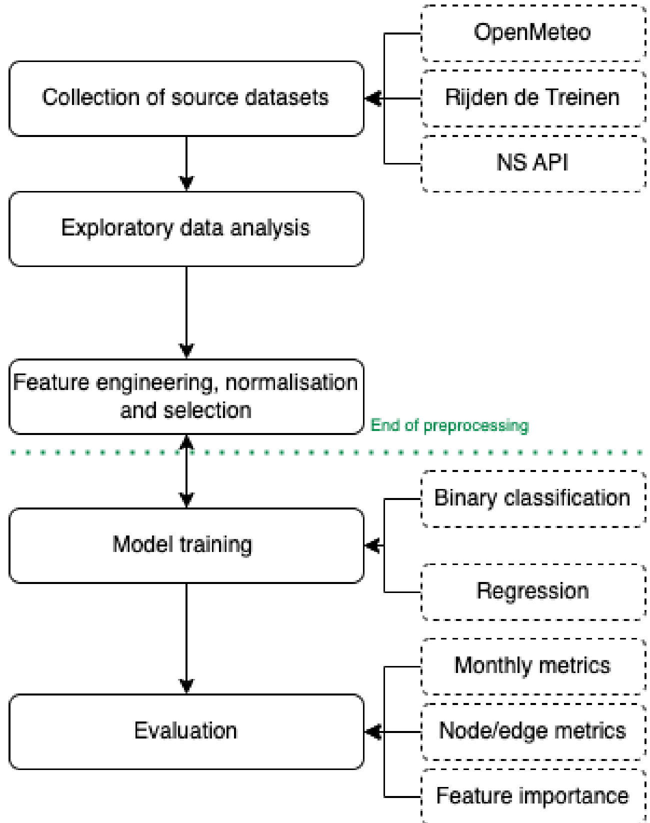


Figure 12: The full project pipeline executed in this research.

C FEATURE CLUSTER CORRELATION

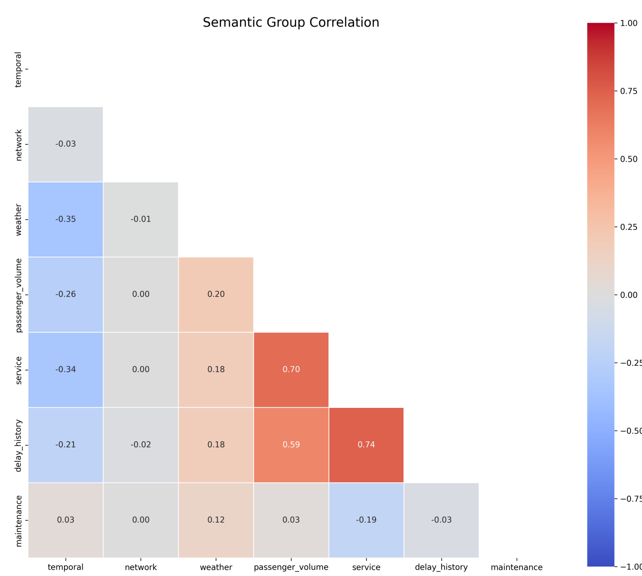


Figure 13: Spearman correlation for semantically grouped features used in the feature importance analysis.

D EDA

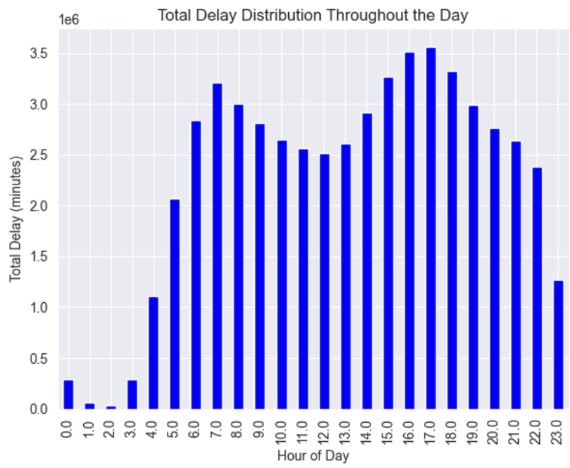


Figure 14: The total arrival delay for each hour in the day. Rush hour peaks are visible for both morning and evening.

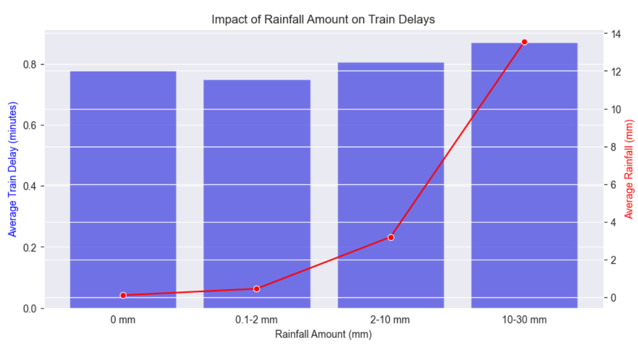


Figure 15: Rainfall vs. the average train arrival delay. The red line shows the average amount of rain within each bucket.

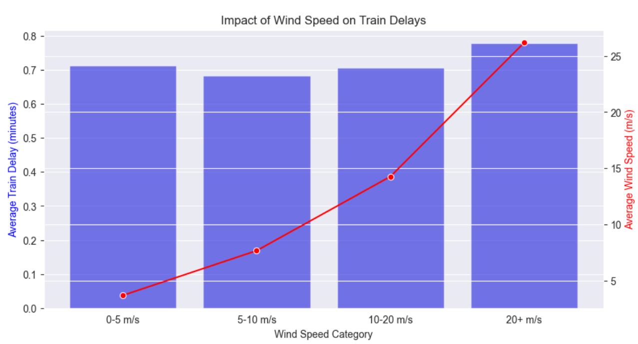


Figure 16: Wind speed vs. the average train arrival delay. The red line shows the average wind speed within each bucket.

E MODEL HYPERPARAMETERS

Model	Hidden Size	Layers	LR	Weight Decay	Seq. Length	Batch Size
GRU	128	3	0.0005	5e-3	10	256
GAT	128	2	0.001	1e-4	1	4
LSTM	128	3	0.0005	5e-3	10	256
GAT-GRU	128	1	0.0005	1e-4	2	1
GAT-LSTM	128	1	0.0005	1e-4	2	1

Table 6: The model hyperparameters for each model trained in this study.

F THE GRU, LSTM AND GAT FORWARD PASS

This section contains additional information on how the GRU, LSTM and GAT make predictions and how this influences the proposed hybrid architectures.

GRU consists of two gates, the reset and update gate. The reset gate r_t decides how much of the previous hidden state will be forgotten when computing the candidate hidden state $\tilde{h}_t$ and the update gate u_t decides how much of the computed candidate hidden state will be used to update the hidden state h_t. At the next timestep h_t becomes h_{t-1} which is how the GRU forward pass retains a memory of previously processed inputs. The GRU input accepts sequences of data which decide how much previous information the model incorporates when making predictions as at each timestep in the sequence, the hidden state is updated and therefore information of the previous timestep is retained [4].

The forward pass for Gated Recurrent Units (GRU) [4].

$$
\begin{aligned}
r_t &= \sigma\left(W_r x_t + U_r h_{t-1} + b_r\right) \\
u_t &= \sigma\left(W_u x_t + U_u h_{t-1} + b_u\right) \\
\tilde{h}_t &= \tanh\left(W_c x_t + U_c (r_t \odot h_{t-1}) + b_c\right) \\
h_t &= u_t \odot h_{t-1} + (1 - u_t) \odot \tilde{h}_t
\end{aligned}
$$

The LSTM works, just as GRU, with gated aggregation in the forward pass. LSTM contains more gates than GRU to tackle the problem of vanishing gradients over longer sequences. Where GRU updates the hidden state with two gates, LSTM utilizes three gates. The input gate, forget gate and output gate. The input gate i_t controls how much of the candidate memory g_t should be added to the cell state c_t, the forget gate f_t determines how much of the previous cell state c_{t-1} should be retained and the output gate determines how much of the cell state should be used in the update of the hidden state. The extra gate compared to GRU improves the LSTM's ability to retain memory over longer periods of time by explicitly controlling when information is exposed to the output through cell state c_t [27].

The forward pass for Long Short-Term Memory (LSTM) [27].

$$
\begin{aligned}
i_t &= \sigma\left(W_i x_t + U_i h_{t-1} + b_i\right) \\
f_t &= \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \\
o_t &= \sigma\left(W_o x_t + U_o h_{t-1} + b_o\right) \\
g_t &= \tanh\left(W_g x_t + U_g h_{t-1} + b_g\right) \\
c_t &= f_t \odot c_{t-1} + i_t \odot g_t \\
h_t &= o_t \odot \tanh(c_t)
\end{aligned}
$$

GAT does not have a gated aggregation mechanism as in GRU and LSTM. GAT works by computing attention weights for a node based on connecting nodes. The equation for e_{ij} shows the computation of the attention weight for edge i, j showing the importance of node j on node i. The second equation α_{ij} normalizes these raw attention weights and in the last equation h_i the attention weights are aggregated to compute the updated hidden state h_i for node i [3].

The forward pass for Graph Attention Networks (GATv2) [3].

$$
\begin{aligned}
e_{ij} &= \text{LeakyReLU}\left(\mathbf{a}^\top \mathbf{W}(x_i + x_j)\right) \\
\alpha_{ij} &= \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik})} \\
h_i &= \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W} x_j\right)
\end{aligned}
$$

The GRU and LSTM architecture have the ability to retain information from previous timesteps by including the previous hidden state in the hidden state computation. GAT does not retain information from previous timesteps but includes weighted updates based on neighboring edges and nodes which leads to an improved ability to include spatial features in hidden state updates. By combining these architectures in a hybrid GAT-GRU and GAT-LSTM cell the proposed hybrid models have an improved ability to incorporate both spatial and temporal dependencies in delay prediction.

G FEATURES USED IN THE EDGE AND NODE DATASETS

Feature Name	Description
Temporal Features	
hour_sin/cos	Cyclical encoded timestamp features
day_of_week_sin/cos	
day_sin/cos	
month_sin/cos	
year_sin/cos	
is_weekend	Weekend indicator
is_rush_hour	Peak travel hours indicator (7AM-9AM and 4PM-6PM)
Network Features	
degree	Number of connections to other stations
distance_from_center	Distance from network center
distance	Distance between connected stations
importance_differential	Difference in station importance based on station type
service_frequency	Average service frequency of the source and target node
source/target_geo_lat/lng_scaled	Station coordinates
Weather Features	
temperature_2m_scaled	Air temperature at 2m
wind_speed_10m_scaled	Wind speed at 10m
wind_gusts_10m_scaled	Wind gust speed at 10m
soil_temperature_0_to_7cm_scaled	Soil temperature at 0-7cm
rain_scaled	Precipitation amount
snowfall_scaled	Snowfall amount
snow_depth_scaled	Snow depth
source/target_temperature	Temperature at source/target station
source/target_wind_speed	Wind speed at source/target station
source/target_wind_gusts	Wind gusts at source/target station
source/target_rain	Precipitation at source/target station
source/target_snowfall	Snowfall at source/target station
source/target_soil_temperature	Soil temperature at source/target station
Passenger and Service Features	
hourly_boarding_deboarding_scaled	Passengers boarding/alighting
hourly_transfers_scaled	Transfer passengers
total_connections_scaled	Connections to other stations
total_services_scaled	Total train services per node-timestamp combination
unique_services_scaled	Unique train services per node-timestamp combination
next_station_x_count_scaled	Number of next stations x steps from the current node (x=1-5)
has_services	Binary flag indicating services at that hour
source/target_boarding_deboarding	Passenger volume at source/target
source/target_transfers	Transfers at source/target
source/target_total_connections_scaled	Total connections at source/target
source/target_total_services_scaled	Services at source/target
source/target_unique_services_scaled	Unique services at source/target
source/target_next_station_x_count_scaled	Next station counts (x=1-3)
Delay History Features	
avg_arrival_delay_lag_x_scaled	Historical arrival delays (x=1-3)
avg_departure_delay_lag_x_scaled	Historical departure delays (x=1-3)
delayed_arrivals_lag_x_scaled	Historical delayed arrivals (x=1-3)
delayed_departures_lag_x_scaled	Historical delayed departures (x=1-3)
cancelled_arrivals_lag_x_scaled	Historical cancelled arrivals (x=1-3)
cancelled_departures_lag_x_scaled	Historical cancelled departures (x=1-3)
cancelled_arrivals/departures_scaled	Current cancelled services
source_delayed_departures	Delayed departures at source (lag of 1 hour)
target_delayed_arrivals	Delayed arrivals at target (lag of 1 hour)
Maintenance and Disruption Features	
expanded_maintenance_count_scaled	Number of concurrent maintenance events
disruption_duration_minutes_scaled	Disruption duration
disruption_duration_minutes_lag_x_scaled	Historical disruption duration (x=1-5)
has_disruption	Current disruption indicator
has_disruption_lag_x	Historical disruption indicators (x=1-5)
has_maintenance_next_xh	Upcoming maintenance (x=3,6,12,24)
source/target_has_disruption	Disruption at source/target
source/target_has_disruption_lag_x	Historical disruptions (x=1-3)
source/target_has_maintenance_next_xh	Upcoming maintenance (x=12,24)
source/target_maintenance	Maintenance at source/target
Target Features	
avg_next_station_x_delay_scaled	Average delay at next stations (x=1-5)
is_significantly_delayed	Binary indicator for significant delay

H STATISTICAL VALIDATION OF RESULTS

Model 1	Model 2	Mean Difference	t	p	Significant (p<0.05)
gat_lstm	gat	0.0468	2.3679	0.0373	Yes
gat_lstm	gru	0.0756	3.4832	0.0051	Yes
gat_lstm	lstm	0.0683	3.3457	0.0065	Yes
gat_lstm	gat_gru	0.0202	7.3804	0.0000	Yes
gat_gru	gat	0.0266	1.2771	0.2279	No
gat_gru	gru	0.0554	2.4342	0.0332	Yes
gat_gru	lstm	0.0481	2.2385	0.0468	Yes

Table 7: Pairwise t-test results to assess classification balanced accuracy results. Monthly balanced accuracy was used as samples (n=12).

Model 1	Model 2	Mean MAE Difference	t	p	Significant (p<0.05)
gat_gru	gat	-0.2136	-1.3611	0.2007	No
gat_gru	gru	-0.0773	-4.8821	0.0005	Yes
gat_gru	lstm	-0.2645	-2.9264	0.0138	Yes
gat_gru	gat_lstm	-0.0061	-2.3211	0.0405	Yes
gat_lstm	gat	-0.2075	-1.3385	0.2077	No
gat_lstm	gru	-0.0712	-4.6909	0.0007	Yes
gat_lstm	lstm	-0.2584	-2.8937	0.0146	Yes

Table 8: Pairwise t-test results to asses MAE results. Monthly average MAE was used as samples (n=12).

Model 1	Model 2	Mean RMSE Difference	t	p	Significant (p<0.05)
gat_gru	gat	-0.2824	-1.3526	0.2033	No
gat_gru	gru	-0.0870	-3.4798	0.0052	Yes
gat_gru	lstm	-0.1284	-1.2276	0.2452	No
gat_gru	gat_lstm	0.0000	0.0252	0.9803	No
gat_lstm	gat	-0.2824	-1.3498	0.2042	No
gat_lstm	gru	-0.0870	-3.4030	0.0059	Yes
gat_lstm	lstm	-0.1285	-1.2235	0.2467	No

Table 9: Pairwise t-test results to asses the RMSE results. Monthly average RMSE was used as samples (n=12).

Model	Coefficient	p	Significant (p<0.05)
GAT-GRU (Avg Degree)	-0.0229	0.0002	Yes
GAT-LSTM (Avg Degree)	-0.0226	4.47×10^{-5}	Yes
GAT-GRU (Source Degree)	-0.0060	0.2108	No
GAT-GRU (Target Degree)	-0.0227	4.57×10^{-6}	Yes
GAT-LSTM (Source Degree)	-0.0076	0.0744	No
GAT-LSTM (Target Degree)	-0.0206	3.31×10^{-6}	Yes

Table 10: Linear regression results between node degree and edge balanced accuracy for both GAT-LSTM and GAT-GRU.

Model	Coefficient	p	Significant (p<0.05)
GAT-GRU (Avg Neighbor Degree)	-0.0179	0.6530	No
GAT-LSTM (Avg Neighbor Degree)	-0.0205	0.6061	No
GAT-GRU (Degree)	0.0988	0.0028	Yes
GAT-LSTM (Degree)	0.0962	0.0034	Yes

Table 11: Linear regression results between node degree and node MAE for both GAT-LSTM and GAT-GRU.

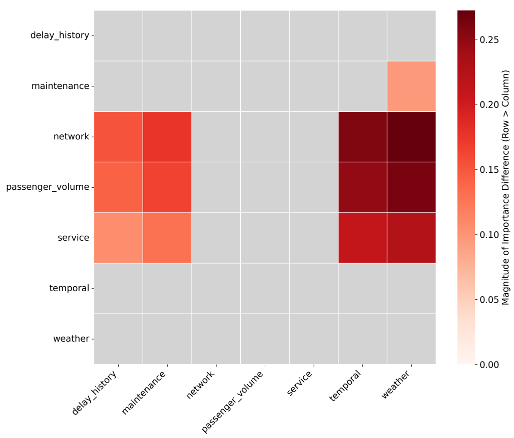


Figure 17: The results of the Wilcoxon signed-rank test for the classification task. Grey blocks are non significant results. The colored blocks mean that the row feature group has a significantly higher Spearman correlation than the column feature.

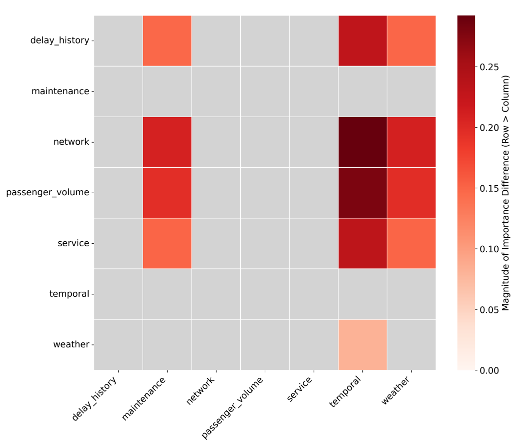


Figure 18: The results of the Wilcoxon signed-rank test for the regression task. Grey blocks are non significant results. The colored blocks mean that the row feature group has a significantly higher Spearman correlation than the column feature.

I ON THE USE OF GENERATIVE AI

There are two main areas where generative AI was used in the process of this master thesis. I was struggling with LaTeX which is one of the two areas where I used GenAI. I especially struggled with properly formatting tables to fit the two column format as they would easily overflow off the page or to the right column so I used GenAI to help out there. Another example is that I used it to put references I found in the right format. I first used mybib.com to generate a Harvard style reference from, for example, a doi or url and then used ChatGPT to put it in the ACM reference format used in the thesis. I did also use the Overleaf language improvement function a few times (less than five) to fix the grammar in a sentence I wrote but apart from that GenAI was not used anywhere in writing the text of my paper.

Apart from that I sometimes used GenAI to help with minor code debugging such as linting errors or indentation problems. I have also used a few times to add debug print statements to my code such as printing tensor shapes so I could find out why I was getting index out of bounds errors. From these prints I learned that the batch size was added to, for example, the node feature tensor. Subsequent code fixes were then done without GenAI. It was the first time for me using GPU acceleration (Snellius). I used GenAI to list what I needed to do in order to actually utilize the GPU in my script such as defining a device variable and moving all node and edge features there for training as I read that simply running a script on a GPU partition does not always properly utilizes the GPU. Again, the actual code for this I wrote myself. I also used it to add some print statement checks to confirm that I was indeed using GPU and not CPU which initially happened to me but that was a result of simply using the wrong Snellius partition. I believe that this use of GenAI is within the bounds of what is allowed as the ideas, text and code in my thesis are mine. I hope I am correct in this assumption.